Quiz Q1. Consider four vectors in $\mathbb{R}^2$:

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} 4 \\ 6 \end{pmatrix}, \quad \mathbf{x}_3 = \begin{pmatrix} 5 \\ 4 \end{pmatrix}, \quad \mathbf{x}_4 = \begin{pmatrix} -2 \\ 3 \end{pmatrix}.$$

Which pair is closest in Euclidean distance?

A. $\mathbf{x}_1$ and $\mathbf{x}_3$

B. $\mathbf{x}_1$ and $\mathbf{x}_4$

C. $\mathbf{x}_2$ and $\mathbf{x}_3$

D. $\mathbf{x}_3$ and $\mathbf{x}_4$

Quiz Q1 answer. The correct answer is C.

$$d(\mathbf{x}_1, \mathbf{x}_2)^2 = 25, \quad d(\mathbf{x}_1, \mathbf{x}_3)^2 = 20, \quad d(\mathbf{x}_1, \mathbf{x}_4)^2 = 10,$$
$$d(\mathbf{x}_2, \mathbf{x}_3)^2 = 5, \quad d(\mathbf{x}_2, \mathbf{x}_4)^2 = 45, \quad d(\mathbf{x}_3, \mathbf{x}_4)^2 = 50.$$

Thus $\mathbf{x}_2$ and $\mathbf{x}_3$ form the closest pair.

Quiz Q2. During a K-Means assignment step, the centroids are

$$\mathbf{c}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{c}_2 = \begin{pmatrix} 5 \\ 4 \end{pmatrix}, \quad \mathbf{c}_3 = \begin{pmatrix} 8 \\ 1 \end{pmatrix}, \quad \mathbf{c}_4 = \begin{pmatrix} 7 \\ 2 \end{pmatrix}.$$

The point $\mathbf{x} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$ originally belongs to $\mathcal{C}_4$. To which cluster is it assigned?

A. $\mathcal{C}_1$

B. $\mathcal{C}_2$

C. $\mathcal{C}_3$

D. $\mathcal{C}_4$

Quiz Q2 answer. The correct answer is D.

$$\|\mathbf{x}-\mathbf{c}_1\|_2^2 = 17, \quad \|\mathbf{x}-\mathbf{c}_2\|_2^2 = 4,$$
$$\|\mathbf{x}-\mathbf{c}_3\|_2^2 = 10, \quad \|\mathbf{x}-\mathbf{c}_4\|_2^2 = 4.$$

The closest centroids are $\mathbf{c}_2$ and $\mathbf{c}_4$. Because $\mathbf{x}$ originally belongs to $\mathcal{C}_4$, the tie rule leaves it in $\mathcal{C}_4$.

Quiz Q3. Two runs of K-Means use the same dataset, the same value of k and the same tie rule, but start from different nonempty initial partitions. Assuming neither run encounters an empty cluster, which statement is correct?

A. They must end with the same clustering because SSE decreases in both runs.

B. They can end with different clusterings.

C. The run with smaller initial SSE must also have smaller final SSE.

D. Both runs must reach the globally smallest possible SSE.

Quiz Q3 answer. The correct answer is B.

Strict decrease of SSE and finite termination apply to each run separately. They do not imply that the final clustering is unique or globally optimal.

Different initial partitions can lead through different sequences of assignments and terminate at different stable clusterings.